INCH-POUND

MIL-PRF-1/480E
1 October 2009
SUPERSEDING
MIL-PRF-1/480D
12 October 1998

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY

TYPES 5SP1A, 5SP7A AND 5SP11A (1/)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus with dual gun.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Type phosphor | Ef | Ec1 | ed | Eb1 | Eb2 | Eb3 | Rg | Zd | Eb3/Eb2 | Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | | V | V dc | V dc | V dc | V dc | V dc | MegΩ | MegΩ | Ratio | ft |
| Maximum: | All | 6.9 | 0 | 600 | 1,100 | 2,750 | 8,250 | 1.5 | 1.0 | 3 | 50,000 |
| Minimum: | All | 5.7 | -200 | - - - | - - - | 1,000 | 1,000 | - - - | - - - | - - - | - - - |
| Test conditions: See note 2 | P1A, P11A<br>P7A | 6.3<br>6.3 | Adj<br>Adj | - - -<br>- - - | Focus<br>Focus | 1,500<br>2,000 | 3,000<br>4,000 | - - -<br>- - - | - - -<br>- - - | - - -<br>- - - | - - -<br>- - - |

GENERAL:

Qualification - Required.

1/ See note 10



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TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| Qualification inspection | | | | | | | |
| Pressure (implosion) | 1141 | All | | - - - | - - - | - - - | - - - |
| Barometric pressure, reduced | 1002 | All | | - - - | - - - | - - - | - - - |
| Vibration | 5111 | All | | Width | - - - | 1 | mm |
| Direct-interelectrode capacitance | 1331 | All | k to all | Ck | - - - | 5.0 | pF |
| | | | g1 to all | Cg1 | - - - | 6.6 | pF |
| | | | D1 to D2 | C1D2 | - - - | 2.0 | pF |
| | | | D3 to D4 | C3D4 | - - - | 2.3 | pF |
| | | | D1 to all except D2 | C | - - - | 2.3 | pF |
| | | | D2 to all except D1 | C | - - - | 3.0 | pF |
| | | | D3 to all except D4 | C | - - - | 3.0 | pF |
| | | | D4 to all except D3 | C | - - - | 2.8 | pF |
| Neck and bulb alignment (electrostatic types) | 5101 | All | | Dia | - - - | 3.5 | inch |
| Cathode illumination | 5216 | All | | - - - | - - - | - - - | - - - |
| Focusing voltage (zero-bias) | 5246 | P1A, P11A | | Eb1 | 272 | 521 | V dc |
| | | P7A | | Eb1 | 363 | 695 | V dc |
| Deflection factor | 5248 | P1A, P11A | 1D2; Eb3 = Eb2 = 1,500 V dc | DF | 49 | 61 | V dc/in. |
| | | P7A | 1D2; Eb3 = Eb2 = 2,000 V dc | DF | 65 | 81 | V dc/in. |
| Deflection factor | 5248 | P1A, P11A | 3D4; Eb3 = Eb2 = 1,500 V dc | DF | 43 | 53 | V dc/in. |
| | | P7A | 3D4; Eb3 = Eb2 = 2,000 V dc | DF | 57 | 71 | V dc/in. |
| Deflection-factor uniformity | 5248 | All | | - - - | - - - | 2 | % |
| Base material insulating quality | 1216 | All | Zone 5, or higher | - - - | - - - | - - - | - - - |

See notes at end of table I.



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TABLE I. Testing and inspection -Continued.

| Inspection | Method MIL-STD-1311 | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| Conformance inspection, part 1 | | | | | | | |
| Voltage breakdown | 5201 | All | | - - - | - - - | - - - | - - - |
| Voltage breakdown (electrostatic types) | 5201 | All | | - - - | - - - | - - - | - - - |
| Gas “cross” | 5206 | P1A<br>P7A<br>P11A | Light = 10 fL (see note 1)<br>Ib3 = 25 μA dc (see note 1)<br>Light = 4.5 fL (see notes 1 and 3) | - - -<br>- - -<br>- - - | - - -<br>- - -<br>- - - | - - -<br>- - -<br>- - - | - - -<br>- - -<br>- - - |
| Bulb, screen and faceplate quality | 5106 | All | | - - - | - - - | - - - | - - - |
| Modulation | 5223 | P1A<br>P7A<br>P11A | Light = 10 fL<br>Ib3 = 25 μA dc<br>Light = 4.5 fL (see note 3) | ΔEc<br>ΔEc<br>ΔEc | - - -<br>- - -<br>- - - | 32<br>42<br>32 | V dc<br>V dc<br>V dc |
| Light output | 5221 | P1A<br>P11A | (see note 1)<br>(see notes 1 and 3) | Light<br>Light | 10<br>4.5 | - - -<br>- - - | fL<br>fL |
| Spot position (electrostatic deflection) | 5231 | All | (see note 5) | - - - | - - - | 16 | mm |
| Spot displacement (leakage) | 5231 | All | | Displ | - - - | 10 | mm |
| Grid-cutoff voltage | 5241 | P1A, P11A<br>P7A | | Ec1<br>Ec1 | -34<br>-45 | -56<br>-75 | V dc<br>V dc |
| Pattern distortion | 5103 | All | (see note 7) | - - - | - - - | - - - | - - - |
| Grid No. 1 leakage current | 5251 | All | | - - - | - - - | - - - | - - - |
| Anode No. 2 leakage current | 5251 | All | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 2 | | | | | | | |
| Secureness of base, cap or insert | 1101 | All | | - - - | - - - | - - - | - - - |
| Heater current | 1301 | All | | If | 540 | 660 | mA |
| Electrode current (anode No. 1) | 5201 | All | Ec1 = 0 | Ib1 | -50 | 10 | μA dc |

See notes at end of table I.



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TABLE I. Testing and inspection -Continued.

| Inspection | Method MIL-STD-1311 | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| Conformance inspection, part 2 -Continued | | | | | | | |
| Electrode current (cathode) | 5201 | P1A<br>P7A<br>P11A | Light = 10 fL<br>Ib3 = 25 μA dc<br>Light = 4.5 fL<br>(see note 3) | Ik<br>Ik<br>Ik | - - -<br>- - -<br>- - - | 1,000<br>1,000<br>1,000 | μA dc<br>μA dc<br>μA dc |
| Base alignment (electrostatic types) | 5101 | All | +4D3, pin No. 4 | - - - | - - - | - - - | - - - |
| Side terminal alignment (electrostatic types) | 5101 | All | +4D3 | - - - | - - - | - - - | - - - |
| Angle between traces | 5101 | All | | - - - | - - - | - - - | - - - |
| Trace alignment | - - - | All | (see note 4) | - - - | - - - | - - - | - - - |
| Neck and base alignment (electrostatic types) | 5101 | All | | - - - | - - - | - - - | - - - |
| Base pin solder depth | 1111 | All | | - - - | - - - | - - - | - - - |
| Stray light emission (conventional types) | 5216 | All | Eb2 = 2,750 V dc<br>Eb3 = 8,250 V dc | - - - | - - - | - - - | - - - |
| Screen | 5221 | P7A | | - - - | - - - | - - - | - - - |
| Line width “A” (electrostatic deflection) | 5226 | P1A<br>P7A<br>P11A | Light = 10 fL<br>Ib3 =25 μA dc<br>Light = 4.5 fL<br>(see note 3) | Width<br>Width<br>Width | - - -<br>- - -<br>- - - | 0.69<br>0.69<br>0.69 | mm<br>mm<br>mm |
| Line width “B” (electrostatic deflection) | 5226 | P1A<br>P7A<br>P11A | Light = 10 fL<br>Ib3 =25 μA dc<br>Light = 4.5 fL<br>(see note 3) | Width<br>Width<br>Width | - - -<br>- - -<br>- - - | 0.69<br>0.69<br>0.69 | mm<br>mm<br>mm |
| Focusing voltage at cutoff | 5246 | P1A, P11A<br>P7A | | Eb1<br>Eb1 | 345<br>460 | 518<br>690 | V dc<br>V dc |
| Deflection factor | 5248 | P1A, P11A<br>P7A | 1D2<br>1D2 | DF<br>DF | 62<br>83 | 76<br>101 | V dc/in.<br>V dc/in. |
| Deflection factor | 5248 | P1A, P11A<br>P7A | 3D4<br>3D4 | DF<br>DF | 53<br>70 | 65<br>86 | V dc/in.<br>V dc/in. |
| Deflection factor ratio | - - - | All | (see note 6) | - - - | - - - | - - - | - - - |

See notes at end of table I.



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TABLE I. Testing and Inspection -Continued.

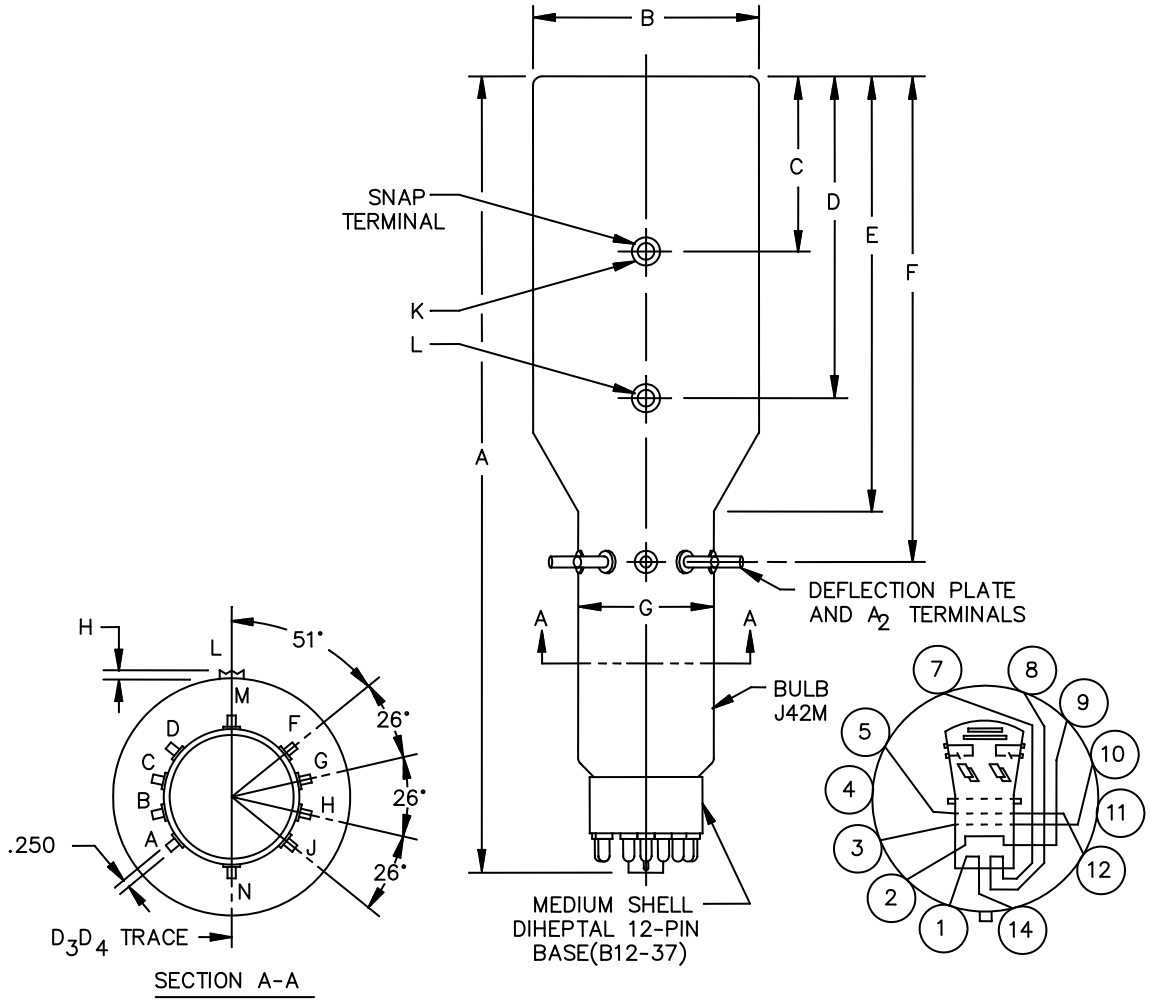
| Inspection | Method MIL-STD-1311 | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| Conformance inspection, part 2 -Continued | | | | | | | |
| Interaction factor | - - - | All | (see note 8) | - - - | - - - | - - - | - - - |
| Heater-cathode leakage current | 5251 | All | | - - - | - - - | - - - | - - - |
| Permanence of marking | 1105 | All | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 3 | | | | | | | |
| Life test | - - - | All | Group C; Ib3 = 15 μA dc; Eb2 = 2,750 V dc; Eb3 = 8,250 V dc | t | 500 | - - - | hrs |
| Life-test end points: | - - - | P1A | Light = 7.5 fL | - - - | - - - | - - - | - - - |
| | | P11A | Light = 3.5 fL (see note 3) | - - - | - - - | - - - | - - - |
| | | P7A | Ib3 = 20 μA dc | - - - | - - - | - - - | - - - |
| Modulation | | P1A, P11A | | ΔEc1 | - - - | 32 | V dc |
| Modulation | | P7A | | ΔEc1 | - - - | 42 | V dc |
| Line width “A” | | All | | Width | - - - | 0.69 | mm |
| Line width “B” | | All | | Width | - - - | 0.69 | mm |

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. All electrical tests are to be made on each gun separately.
3. As measured with a 2 inch x 2 inch (50.8 x 50.8 mm) raster using type 3 photronic cell without eye correction, calibrated in footcandles of illumination from a light source having a color temperature of 2700°K.
4. Corresponding traces of each gun should be within 1 degree of each other.
5. Under stable operating conditions (method 5231), the position of either spot will not shift with changes in intensity by more than 0.025 inch (0.64 mm).
6. Ratio of deflection factors of corresponding sets of deflection plates (larger deflection factor divided by lower deflection factor) shall not exceed 1.15.
7. With a raster pattern the size of which is adjusted so that the widest points of the pattern just touch the sides of a square, 3.075 inches (78.11 mm) on a side, no point on these pattern sides will lie within an inscribed square, 2.925 inches (74.30 mm) on a side.
8. The deflection of one beam when balanced dc voltages are applied to the deflection electrodes of the other beam will not be greater than the indicated value.
9. Maximum voltages applied to base, deflection plates and A2 only.
10. Tube type 5SP2A has been deleted from this specification sheet.
11. This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.



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Connections

Terminal K Intensifier Electrode
Terminal L Accelerating Electrode Coating

| | Unit A | | Unit B | |
|---|---|---|---|---|
| Pin No. | 1. | Heater | 7. | Heater |
| | 2. | Cathode | 8. | Heater |
| | 3. | Control Electrode | 9. | Cathode |
| | 4. | Internal connection | 10. | Control Electrode |
| | 5. | Focusing Electrode | 11. | No connection |
| | 14. | Heater | 12. | Focusing Electrode |
| Terminal | A | Deflection Plate D2 | F | Deflection Plate D1 |
| | B | Deflection Plate D1 | G | Deflection Plate D2 |
| | C | Deflection Plate D3 | H | Deflection Plate D4 |
| | D | Deflection Plate D4 | J | Deflection Plate D3 |
| | M | Accelerating Electrode | N | Accelerating Electrode |

FIGURE 1. Outline drawing of electron tube types 5SP1A, 5SP7A, and 5SP11A.



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| Dimensions in inches with metric equivalents (mm) in parentheses | | |
|---|---|---|
| Ltr | Minimum | Maximum |
| Conformance inspection, part 2 | | |
| A | 18.000 (457.20) | 18.500 (469.90) |
| B | 5.156 (130.96) | 5.344 (135.74) |
| C | 3.750 (95.25) | 4.250 (107.95) |
| D | 7.000 (177.80) | 7.500 (190.50) |
| F | 10.750 (273.05) | 11.250 (285.75) |
| G | 2.937 (74.60) | 3.062 (77.77) |
| Conformance inspection, part 3 (periodic check) | | |
| BULB: J42M | | |
| BASE: B12-37 | | |
| Reference dimensions (see note c) | | |
| E | 10.000 (254.00) | |
| H | 0.125 (3.18) MAX | |

NOTES:

1. M-N connected together internally.
2. L and M-N shall be connected together to a common ground.
3. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 5SP1A, 5SP7A and 5SP11A - Continued.

Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5960-2009-010)

Review activities:
Navy - AS, CG, MC, OS
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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